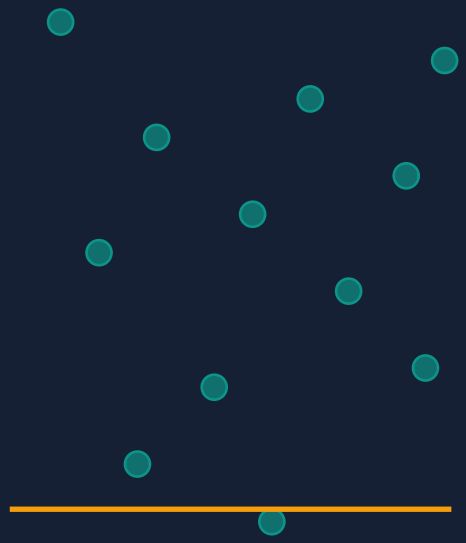
# Linear Regression

## Theory, Assumptions & RStudio

*Based on Andy Field – Discovering Statistics Using R*



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# What is Linear Regression?

*"A statistical method that models the relationship between one or more predictor variables and a continuous outcome variable by fitting a straight line."*

*For prediction and control for confounder , quantifying the relationship -Simple and multiple (crude and adjusted)*

| Outcome (Y) | Predictor (X) | The Line |
|---|---|---|
| The variable we want to predict or explain (dependent variable) | The variable(s) used to predict the outcome (independent variable) | Best-fit line minimising the sum of squared residuals (OLS method)<br><br>Other method<br>Maximum Likelihood Estimation (MLE) for logistic regression |

**The Regression Equation:**

$$\hat{Y}_i = b_0 + b_1 X_i + \varepsilon_i$$

*$b_0$ = intercept  $b_1$ = slope/coefficient  $\varepsilon_i$ = error/residual*

*Field (2013): Discovering Statistics Using R, p. 183*

# Theory: How OLS Works

## Ordinary Least Squares (OLS)

- Each data point has a residual: the vertical distance between the observed (Y) and predicted ($\hat{Y}$) value
- OLS finds the line that minimises the Sum of Squared Residuals (SSR)
- $SSR = \Sigma(Y_i - \hat{Y}_i)^2 \rightarrow$ minimised
- Squaring ensures negative and positive errors don't cancel out
- The resulting $b_0$ and $b_1$ are the 'best' linear estimates

## Model Fit: $R^2$ & F-ratio

### $R^2$

Proportion of variance in Y explained by the model. $R^2 = 0.60 \rightarrow$ model explains 60% of variance in Y.

### Adjusted $R^2$

Penalises for extra predictors. Preferred when comparing models (Field, 2013).

### F-ratio

Tests whether the model significantly improves prediction over the mean alone. $F = MS_m / MS_r$

### β (Beta)

Standardised coefficients allow comparison of predictor importance across different scales.

*Field (2013), pp. 184–205*

# Assumptions of Linear Regression

1

## Linearity

Relationship between X and Y must be linear. Check: scatterplot of X vs Y; residuals vs fitted plot.

2

## Independence

Observations must be independent of each other. Check: study design; Durbin-Watson test (≈2 = no autocorrelation).

3

## Homoscedasticity

Residuals should have constant variance across levels of X. Check: residuals vs fitted plot — no funnel shape.

4

## Normality of Residuals

Residuals should be normally distributed. Check: Q-Q plot, histogram of residuals, Shapiro-Wilk test.

5

## No Multicollinearity

Predictors should not be too highly correlated (multiple regression). Check: VIF < 10; Tolerance > 0.1 (Field, 2013).

6

## No Influential Outliers

Extreme cases unduly influence the model. Check: Cook's Distance > 1 suggests influential cases (Field, 2013).

*Field (2013), Chapter 7 – All assumptions must be met for valid inference.*

# Running Linear Regression in RStudio (Step 1–3)

## STEP 01

### Load & Inspect Your Data

```
# Load data
library(tidyverse)
data <- read.csv("mydata.csv")

# Inspect
head(data)
str(data)
summary(data)
```

*Ensure outcome is continuous and predictors are correctly coded.*

## STEP 02

### Visualise the Relationship

```
# Scatterplot
ggplot(data, aes(x = predictor,
                 y = outcome)) +
  geom_point(colour = "steelblue") +
  geom_smooth(method = "lm",
              se = TRUE) +
  theme_minimal()
```

*Look for linear trend before fitting the model.*

**Assumption 1**
**Linearity**

## STEP 03

### Fit the Regression Model

```
# Simple linear regression
model <- lm(outcome ~ predictor,
            data = data)

# View full output
summary(model)
```

*lm() = linear model. Outcome ~ Predictor syntax follows R formula notation.*

*Field (2013), pp. 211–220 | Code adapted for R / tidyverse*

# Interpreting the summary() Output

```
Coefficients:

            Estimate  Std.Error  t value   Pr(>|t|)
(Intercept)  10.543     1.234     8.543    <0.001 ***
predictor     0.621     0.087     7.138    <0.001 ***

---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05

Residual standard error: 3.21 on 98 df
Multiple R-squared:  0.612
Adjusted R-squared:  0.608
F-statistic: 51.95 on 1 and 98 DF
p-value: < 2.2e-16
```

**Intercept ($b_0$)**

Predicted Y when X = 0

**Slope ($b_1$ = 0.621)**

For each 1-unit ↑ in X, Y increases by 0.621

**p-value < .001**

Predictor is statistically significant

**$R^2$ = 0.612**

Model explains 61.2% of variance in outcome

**F-statistic**

Overall model is significant ($p < .001$)

*Field (2013), pp. 221–235*

# Checking Assumptions in RStudio (Step 4)

## What to Look For:

```
# ── Dubin watson test ──────────────────
dwtest(model)

# ── Diagnostic Plots ──────────────────
# Base R: 4 plots in one window
par(mfrow = c(2,2))
plot(model)

# ── Homoscedasticity ──────────────────
library(lmtest)
bptest(model)  # Breusch-Pagan test

# ── Normality of Residuals ────────────
shapiro.test(residuals(model))

# ── Multicollinearity (if multiple X) ──
library(car)
vif(model)       # VIF < 10 = acceptable
tolerance <- 1/vif(model)  # > 0.1

# ── Influential Cases ─────────────────
cooksD <- cooks.distance(model)
plot(cooksD, type="h")
abline(h = 1, col="red", lty=2)
```

### Independence

Dubin Watson test – p value not significant and value ~ 2 (Assumption 2)

### Residuals vs Fitted

Random scatter around zero → linearity & homoscedasticity OK (Assumption 3)

### Scale-Location

Horizontal band → constant variance (homoscedasticity) (Assumption 3) p value should be not significant

### Normal Q-Q Plot

Points on diagonal line → residuals are normally distributed (Assumption 4)

### VIF / Tolerance

VIF < 10 and Tolerance > 0.1 → no multicollinearity (Assumption 5)

### Cook's Distance

No points above 1 → no unduly influential cases (Assumption 6)

*Field (2013), pp. 236–255 | car package by Fox & Weisberg (2019)*

# VIOLATION 1 LINEARITY

## SITUATION

*Look visually at scatter plot and residual plot*
*Scatter plot – DV vs IV*
*Residual plot – residual vs fitted values*

*Either transform*

*Or use non-linear regression*

## WHAT CAN YOU DO

- ✓ Use polynomial regression
- ✓ Use spline regression
- ✓ Use transformation (i.e log / square root)

```
poly_model <- lm(y ~ poly(x, 2))
plot(x, fitted(poly_model))
spline_model <- lm(y ~ ns(x, df=5))
plot(x, fitted(spline_model))
y_log <- log(y)
y_sqrt <- sqrt(y)
```

*Field (2013), pp. 255–260 | APA Publication Manual, 7th Ed.*

# VIOLATION 2 INDEPENDENCE / AUTOCORRELATION

## SITUATION

*P Value significant or Dubin Watson value not ~ 2*

Suggesting time series

Use time series model – ARIMA

## WHAT CAN YOU DO

✓ Use time series model

ARIMA

*Field (2013), pp. 255–260 | APA Publication Manual, 7th Ed.*

# VIOLATION 3 Homoscedasticity

## SITUATION

residuals vs fitted plot — no funnel shape
*Or Breusch pagan test p value significant*

*Either use robust standard error*

*Or Transform data*

*Or use Weighted Least Square (WLS) Method*

## WHAT CAN YOU DO

- ✓ Use robust standard error - Use the sandwich package to calculate heteroscedasticity-consistent (HC) standard errors and lmtest for testing.
  coeftest(model, vcov = vcovHC(model, type = "HC3"))
- ✓ Transform data by log transform the outcome - model_log <- lm(log(y) ~ x, data = dataset)
- ✓ WLS Method

*Field (2013), pp. 255–260 | APA Publication Manual, 7th Ed.*

# VIOLATION 4 Residual normality

**SITUATION**

Q-Q plot not normal

If sample size is big , result is still robust

Check for outliers as well and remove them

Either transform the dependent variable

Or use robust regression

## WHAT CAN YOU DO

✓ Transform the DV - model2 <- lm(log(y) ~ x, data = df)

✓ Robust regression - library(robustbase); robust_model <- lmrob(y ~ x, data = df)

*Field (2013), pp. 255–260 | APA Publication Manual, 7th Ed.*

# VIOLATION 5 MULTICOLLINEARITY

**SITUATION**

2 or more predictor variables in a regression model are highly correlated, leading to unreliable estimates of the coefficients

Either drop one and re run

Or Ridge regression is a method by which we add a degree of bias to the regression estimates.

**WHAT CAN YOU DO**

- ✓ Drop the highly correlated predictors
- ✓ Regularization (ridge/lasso regression)

Repeat model without one of the correlated predictors

```
ridge_model <- glmnet(as.matrix(cbind(x1, x2, x3)), y, alpha=0)
summary(ridge_model)
plot(ridge_model)
coef(ridge_model, s = 0.1)
```

*Field (2013), pp. 255–260 | APA Publication Manual, 7th Ed.*

# VIOLATION 6 OUTLIERS

**SITUATION**
Cook distance – if value >1 or 4/n

Check data entry first

Or data transformation of the dependent variable

Or delete the outliers

Or using robust regression

## WHAT CAN YOU DO

Robust regression-
rlm() (MASS package): Uses Iteratively Reweighted Least Squares (IRLS).

```
library(MASS)
robust_model <- rlm(y ~ x, data = mydata)
summary(robust_model)
```

*Field (2013), pp. 255–260 | APA Publication Manual, 7th Ed.*

# Variable Selection Methods

*Choosing which predictors to include — four common strategies*

## Enter (Forced Entry)

✓ **Pros**

No data dredging; theory-driven; preferred in epidemiology

✗ **Cons**

Relies on researcher expertise; may include non-significant vars

**When to use:**

*Confirmatory research with a priori hypotheses*

```
model <- lm(y ~ x1 + x2 + x3,
data=df)
summary(model)
```

## Forward Selection

✓ **Pros**

Adds only significant variables; efficient for large datasets

✗ **Cons**

Misses suppressor variables; order-dependent; inflated α

**When to use:**

*Exploratory analysis with many candidate predictors*

```
library(MASS)
step(lm(y~1,data=df),
  scope=~x1+x2+x3,
  direction="forward")
```

## Backward Elimination

✓ **Pros**

Starts with full model; better for correlated predictors

✗ **Cons**

Computationally intensive; same α inflation issue

**When to use:**

*When you start with a plausible full model*

```
full <- lm(y~x1+x2+x3,
data=df)
step(full,
  direction="backward")
```

## Stepwise (Both)

✓ **Pros**

Most flexible; re-evaluates at each step

✗ **Cons**

Highest risk of over-fitting; not recommended for inference

**When to use:**

*Prediction-focused (not causal) models only*

```
step(full,
  direction="both")
```

⚠ Automatic selection methods inflate Type I error and should not be used for causal inference — prefer Enter method in aetiological studies.

# Comparing Selection Methods at a Glance

*Summary table for quick reference*

| Method | Direction | Starting point | Criterion | Best use | Epidemiological use |
|---|---|---|---|---|---|
| **Enter** | All at once | None | Theory / a priori | Confirmatory research | ✓ **Recommended** |
| **Forward** | Add predictors | Null model | Lowest p-value | Screening, prediction | ⚠ **Caution** |
| **Backward** | Remove predictors | Full model | Highest p-value | Moderate # predictors | ⚠ **Caution** |
| **Stepwise** | Both directions | Any | AIC or p-value | Prediction only | ✗ **Not recommended** |

*AIC (Akaike Information Criterion) — lower AIC = better model. Use AIC for automated selection rather than p-values to reduce multiple testing bias.*

```
# Use AIC-based stepwise (preferred over p-value based)
library(MASS)
step(full_model, direction = "both", k = 2)   # k=2 = AIC

# Compare models with AIC
AIC(model1, model2, model3)

# BIC (stricter penalty for complexity)
BIC(model1, model2)
```

# Assessing Model Fit

*Use multiple metrics together — no single statistic tells the whole story*

$R^2$

### $R^2$ (R-squared)

Proportion of variance in Y explained by model. $R^2$=0.60 means 60% explained.
Range: 0–1; higher = better fit. Increases with every added variable.

$R^2adj$

### Adjusted $R^2$

Penalises for adding non-informative predictors. Use this for model comparison.
Decreases if added variable doesn't improve fit meaningfully.

F

### F-statistic & p-value

Tests $H_0$: all $\beta = 0$ (null model). Significant F means model explains variance better than intercept-only.

AIC

### AIC / BIC

Information criteria for comparing non-nested models. Lower = better. BIC applies a stricter complexity penalty.

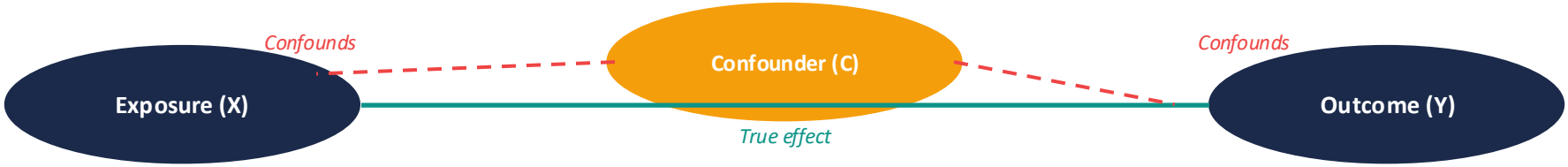
√MSE

### RMSE

Root Mean Square Error — average prediction error in Y units. Useful for comparing predictive accuracy across models.

# Controlling for Confounders

*Multivariable regression adjusts for variables that distort the exposure–outcome relationship*




## How Multivariable Regression Controls Confounding

### 1

**Include confounders as covariates**

Add suspected confounders (age, sex, BMI…) as additional predictors in the model

### 2

**Interpretation: Adjusted coefficient**

β for exposure = effect of X holding all other variables constant

### 3

**Check % change in coefficient**

If adding confounder changes β by >10%, it was confounding the association

### 4

**Report adjusted vs unadjusted**

Always show both crude and adjusted estimates to demonstrate confounding

# Running Linear Regression in RStudio

*The lm() function fits ordinary least squares regression*

```
# Simple (unadjusted) model
model1 <- lm(sbp ~ smoking, data = mydata)
summary(model1)

# Multivariable (adjusted) model
model2 <- lm(sbp ~ smoking + age + bmi + sex,
             data = mydata)
summary(model2)

# Confidence intervals
confint(model2)

# Compare crude vs adjusted
coef(model1)["smoking"]   # crude β
coef(model2)["smoking"]   # adjusted β

# % change (confounding check)
(coef(model1)[2] - coef(model2)[2]) /
  coef(model1)[2] * 100
```

## Reading the summary() Output

**`Estimate`**
Regression coefficient (β) — change in Y per unit X

**`Std. Error`**
Precision of the estimate; smaller = more precise

**`t value`**
Estimate ÷ SE; tests whether β differs from zero

**`Pr(>|t|)`**
p-value; <0.05 = statistically significant

**`R-squared`**
Proportion of variance in Y explained by model

**`F-statistic`**
Tests overall model significance

# Linear regression - Moderator

To test moderation, we include an **interaction term**:

$$Y = \beta_0 + \beta_1 X + \beta_2 M + \beta_3 (X \times M) + \epsilon$$

If **β3 is statistically significant**, moderation exists.

## Check and visualize

```
model <- lm(Outcome ~ Predictor * Moderator, data = mydata)
summary(model)

library(interactions)
interact_plot(model, pred = Predictor, modx = Moderator)
```

## Interpret

use sim_slopes from the interactions package
(model, pred = Predictor, modx = Moderator)

examine the effect of on at different levels of
(e.g., SD)

## Interaction level

Categorical and categorical
Categorical and numeric
Numeric and numeric

*Field (2013): Discovering Statistics Using R, p. 183*

# Linear regression – Mediator

Baron & Kenny Regression Approach (Classic Method)
In this classic method, mediation is tested using **three regression models**.

### Step 1: Total Effect

$Y=\beta 0+\beta 1X$
If $\beta 1$ is significant → X affects Y.

### Step 2: X → Mediator

$M=\beta 0+\beta 1X$
If significant → X influences mediator.

### Step 3: Mediator → Outcome

$Y=\beta 0+\beta 1X+\beta 2M$
Interpretation:
**β2 significant** → mediator affects Y
If **β1 decreases** compared to Step 1 → mediation exists

**The Regression Equation:**

*Full mediation* - Effect of X on Y disappears after mediator added
*Full mediation* - Effect of X decreases but remains significant

*Field (2013): Discovering Statistics Using R, p. 183*

# Mediator – modern approach using indirect effect

Modern mediation uses **indirect effect**:

$$Indirect\ Effect = a \times b$$

Where:
**a** = effect of X → M
**b** = effect of M → Y

Significance usually tested using:
**Bootstrapping (preferred)**
Sobel test

## WHAT CAN YOU DO

- ✓ Bootstrapping - bootstrapping for mediation analysis is commonly done using packages such as mediation, lavaan, or boot. The easiest and most commonly used approach is the mediation package.
- ✓ Sobel test , using BDA package – significant p value indicated mediator

*Field (2013), pp. 255–260 | APA Publication Manual, 7th Ed.*

# Reporting Results – APA Style (Field, 2013)

### Example Write-Up

*A simple linear regression was conducted to predict exam scores from revision hours. The assumptions of linearity, independence, homoscedasticity, and normality of residuals were all met. The model was statistically significant, $F(1, 98) = 51.95$, $p < .001$, $R^2 = .61$. Revision hours significantly predicted exam scores, $b = 0.62$, $t(98) = 7.14$, $p < .001$. For every additional hour of revision, exam scores increased by 0.62 marks on average.*

## Reporting Checklist

- ✓ State assumptions checked and results
- ✓ Report F-statistic, df, and p-value for overall model
- ✓ Report $R^2$ (and Adjusted $R^2$ for multiple regression)
- ✓ Report b (unstandardised) and β (standardised) coefficients
- ✓ Report t-value and p-value for each predictor
- ✓ Describe direction and magnitude of the effect in plain language

*Field (2013), pp. 255–260 | APA Publication Manual, 7th Ed.*

# Regression table

Beta coefficient / OR

Standard error

t statistics

*p* value

Confidence interval

confidence intervals in separate columns

**Table 3**

*Moderator Analysis: Types of Measurement and Study Year*

| Effect | Estimate | *SE* | 95% CI | | *p* |
|---|---|---|---|---|---|
| | | | *LL* | *UL* | |
| Fixed effects | | | | | |
| Intercept | .119 | .040 | .041 | .198 | .003 |
| Creativity measurement[a] | .097 | .028 | .042 | .153 | .001 |
| Academic achievement measurement[b] | –.039 | .018 | –.074 | –.004 | .03 |
| Study year[c] | .0002 | .001 | –.001 | .002 | .76 |
| Goal[d] | –.003 | .029 | –.060 | .054 | .91 |
| Published[e] | .054 | .030 | –.005 | .114 | .07 |
| Random effects | | | | | |
| Within-study variance | .009 | .001 | .008 | .011 | <.001 |
| Between-study variance | .018 | .003 | .012 | .023 | <.001 |

*Note*. Number of studies = 120, number of effects = 782, total *N* = 52,578. CI = confidence interval; *LL* = lower limit; *UL* = upper limit.

[a]0 = self-report, 1 = test. [b]0 = test, 1 = grade point average. [c]Study year was grand centered.

[d]0 = other, 1 = yes. [e]0 = no, 1 = yes.

**Table 7.16 Sample Regression Table, With Confidence Intervals in Brackets**

**Table 4**

*Regressions of Associations Between Marital Satisfaction and Average Levels of Marital Behavior*

| Variable | *B* | *SE* | *t* | *p* | 95% CI |
|---|---|---|---|---|---|
| | | | Angry behavior | | |
| Actor | | | | | |
| H → H | −98.90 | 40.20 | −2.46 | .016 | [− 179.1, −18.7] |
| W → W | −.87.11 | 30.87 | −2.82 | .006 | [−148.7, −25.6] |
| Partner | | | | | |
| W → H | −76.18 | 39.43 | −1.93 | .057 | [−154.8, 2.4] |
| H → W | −91.80 | 38.16 | −2.41 | .019 | [−167.9, −15.7] |
| | | | Disregard | | |
| Actor | | | | | |
| H → H | −38.62 | 27.86 | −1.39 | .170 | [−94.2, 16.9] |
| W → W | −47.54 | 26.99 | −1.76 | .082 | [−101.4, 6.3] |
| Partner | | | | | |
| W → H | −82.81 | 32.01 | −2.59 | .012 | [−146.6, −19.0] |
| H → W | −79.36 | 27.16 | −2.92 | .005 | [−133.5, −25.2] |
| | | | Distancing | | |
| Actor | | | | | |
| H → H | −47.42 | 24.72 | −1.92 | .059 | [−96.7, 1.9] |
| W → W | 3.04 | 23.48 | 0.13 | .897 | [−43.8, 49.8] |
| Partner | | | | | |
| W → H | −0.05 | 23.91 | 0.00 | .998 | [−47.7, 47.6] |
| H → W | −53.50 | 24.47 | −2.19 | .032 | [−102.3, −4.7] |

*Note.* CI = confidence interval; H → H = husband-as-actor effect on the husband's own marital satisfaction; W → W = wife-as-actor effect on the wife's own marital satisfaction; W → H = wife-as-partner effect on the husband's satisfaction; H → W = husband-as-partner effect on the wife's satisfaction.

square brackets around confidence intervals

# Summary & Key Takeaways

1. Linear regression models a straight-line relationship between X and Y
2. OLS minimises the Sum of Squared Residuals to find the best-fit line
3. Always check the 6 assumptions before interpreting results
4. Use lm() in R; plot(model) gives 4 essential diagnostic plots
5. Report F, $R^2$, b coefficients, t-values and p-values in APA style
6. Field (2013) recommends checking Cook's D > 1 for influential cases

*Questions & Discussion*

**Key Reference**

*Field, A. (2013).*
*Discovering Statistics Using R.*
*SAGE Publications.*

*Chapters 7–8*
*pp. 183–260*